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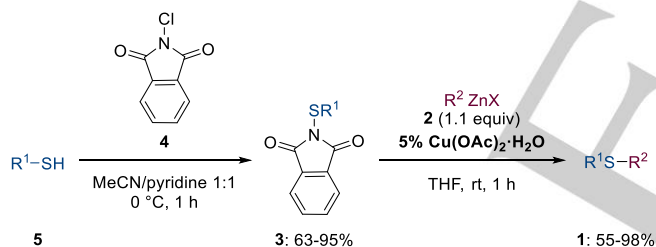
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Copper-Catalyzed Electrophilic Thiolation of Organozinc Halides Using *N*-Thiophthalimides Leading to Polyfunctional Thioethers

Simon Graßl, Clémence Hamze, Thaddäus J. Koller and Paul Knochel*

Abstract: (Hetero)aryl, benzylic and alkyl zinc halides were thiolated with *N*-thiophthalimides at 25 °C within 1 h in the presence of 5–10% Cu(OAc)₂·H₂O to furnish the corresponding polyfunctionalized thioethers in good yields. This electrophilic thiolation was extended to the introduction of trifluoromethylthio (SCF₃), thiocyanate (SCN) and selenophenyl (SePh) groups. The utility of this method was shown in a seven step synthesis of a potent cathepsin D inhibitor in 34% overall yield.

Thioethers are very common functional group in bioactive molecules^[1] and the synthesis of the sulphur linkage led to the development of numerous methodologies.^[2] Thus, Cu-catalyzed nucleophilic thiolations using aryl halides and organic thiols have been reported.^[3] Also, electrophilic thiolations involving electrophilic sulfur-reagents of type RS-X (X = SR, halogeno, *N*-succinimidyl and arylsulfonyl) with various organometallics have found growing interest.^[4] In this respect, the use of polyfunctional organozinc reagents^[5] is of special interest, since they are readily prepared and tolerate a broad range of functional groups like an ester, a nitrile or an aldehyde.



Scheme 1. General preparation of *N*-thiophthalimides **3** and their reaction with organozinc reagents **2** to provide thioethers of type **1**.

Herein, we wish to report a convenient preparation of polyfunctional thioesters of type **1** by the reaction of functionalized aryl, heteroaryl, alkyl or benzylic organozinc halides **2** with *N*-thiophthalimides **3**. These *N*-thiophthalimides were conveniently prepared by an optimized literature procedure based on the methods published by Liebeskind^[4a] and Chen^[6], involving the reaction of commercial *N*-chlorophthalimide **4** with various organic thiols of type **5** (Scheme 1).

In preliminary experiments, we have examined the effect of a transition metal catalyst on the formation of thioether **1a**. Therefore, 4-anisylzinc chloride (**2a**)^[7] was treated with *N*-(cyclohexylthio)phthalimide (**3a**; 25 °C, 1 h). In the absence of a catalyst, a yield of only 49% of **1a** was obtained (Table 1, entry 1). Catalysis with FeCl₂, CrCl₂, Fe(acac)₃, CoCl₂, NiCl₂ and MnCl₂ (5%) increased this yield up to 72% (entries 2–7). By testing various copper salts such as CuBr₂, CuBr, CuCl₂, Cu(OAc) and CuCN·2LiCl, we have further increased the yield up to 88% (entries 8–12). However, CuCl and Cu(OAc)₂·H₂O were found to give the best results (93–95% yield, entries 13–14).

Table 1. Catalyst screening for the electrophilic thiolation of zinc organometallic **2a**.

Entry	Catalyst	Yield [%] ^[a]	Entry	Catalyst	Yield [%] ^[a]
1	-	49	8	CuBr ₂	74
2	FeCl ₂	54	9	CuBr	79
3	CrCl ₂	60	10	CuCl ₂	82
4	Fe(acac) ₃	63	11	Cu(OAc)	88
5	CoCl ₂	64	12	CuCN·2LiCl	88
6	NiCl ₂	71	13	CuCl	93
7	MnCl ₂	72	14	Cu(OAc) ₂ ·H ₂ O	95 ^[b]

^[a] GC yield using undecane as internal standard ^[b] Isolated yields of analytically pure products

With this optimized procedure, we have examined the scope of organozinc reagents **2**. Thus, the reaction of electron-rich (**2a–e**) as well as electron-poor arylzinc chlorides (**2f–g**) with *N*-thiophthalimide **3a** or **3b** using 5% Cu(OAc)₂·H₂O provided the desired thioethers **1b–h** in 86–99% yield (Table 2, entries 1–7). Moreover, ferrocenylzinc chloride (**2h**) was coupled with thiophthalimide **3b**, furnishing the ferrocenyl thioether **1i** in 90% yield (entry 8). Also heteroarylzinc halides **2h–k** underwent an electrophilic thiolation giving the thioethers **1j–l** in 71–94% yield (entries 9–11). Furthermore, alkyl and benzylic zinc halides **2l–n** were successfully coupled with *N*-(phenylthio)phthalimide (**3b**) and *N*-(cyclohexylthio)phthalimide (**3a**), furnishing the thioethers **1m–o** (entries 12–14). Noteworthy, sensitive functional groups such as nitriles, esters, aldehydes and ketones were tolerated (entries 6, 7, 9, 10 and 14). Also, sterically hindered nucleophiles such as zinc organometallics **2c** and **2k** gave the desired thioethers **1d** and **1l** in 94–99% yield (entry 5 and 11).

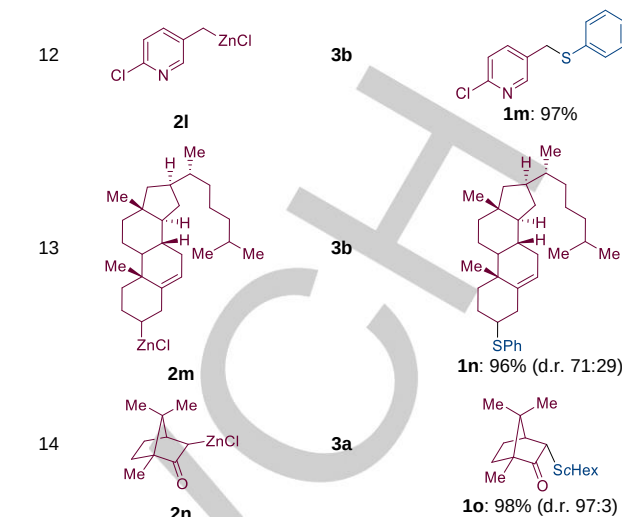
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Table 2. Thioethers of type **1** via a copper-catalyzed thiolation of organozinc reagents of type **2** with *N*-thiophthalimides of type **3**.

$\text{3} + \text{R}^1\text{-ZnCl} \xrightarrow[\text{THF, rt, 1 h}]{5\% \text{ Cu(OAc)}_2 \cdot \text{H}_2\text{O}} \text{R}^1\text{-SR}^2$			
Entry	Zinc reagent 2	Pht-SR 3 ^[a]	Product of type 1 ^[b]
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			



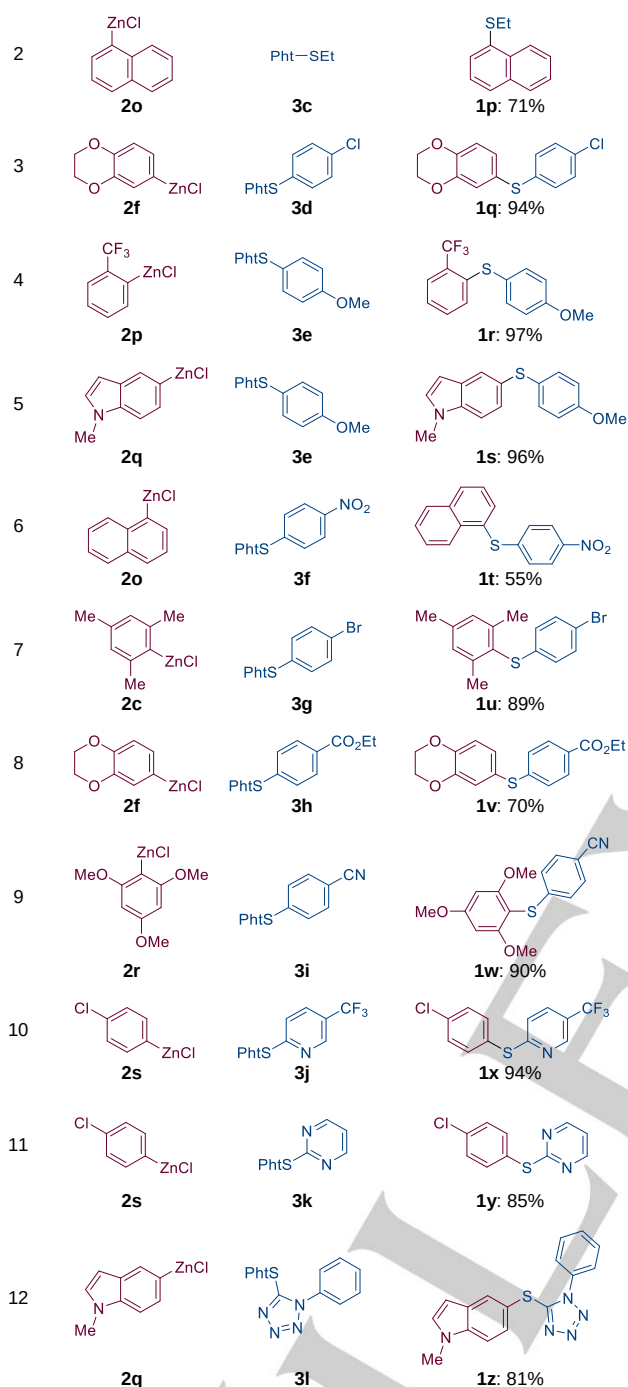
[a] Pht = phthalimide; [b] Isolated yields of analytically pure products.

Next, we have investigated the scope of polyfunctional thiophthalimides of type **3**. We have found that both, alkyl and (hetero)aryl thiophthalimides gave excellent results. Therefore, the alkyl thiophthalimides **3a** and **3c** were coupled with arylzinc chlorides **2a** and **2o** affording thioethers **1a** and **1p** in 71-95% yield (Table 3, entries 1-2). Also various functionalized aryl- and heteroaryl thiophthalimides have been prepared according to the method described above (Scheme 1) and subsequently coupled to form the corresponding thioether of type **1**. Thus, the copper-catalyzed thiolation of organozinc reagents **2c**, **2f**, **2p-s** and thiophthalimides **3d-l** led to the expected thiolation products **1p-y** in 55-97% yield (entries 3-12). Electron-rich as well as electron-poor aryl and heteroaryl derivatives can be used for this thiolation. Furthermore, sensitive functional groups such as esters and nitriles attached to the *N*-thiophthalimide were tolerated (entries 8-9). Noteworthy, even highly reactive nitro groups are tolerated with only slightly reduced yields. Thus, the reaction of thiophthalimide **3f** with organozinc chloride **2o** provided thioether **1s** in 55% yield (entry 6).

Table 3: Electrophile scope of electrophilic thioether formation using various *N*-thiophthalimides of type **3**.

$\text{3} + \text{R}^1\text{-ZnCl} \xrightarrow[\text{THF, rt, 1 h}]{5\% \text{ Cu(OAc)}_2 \cdot \text{H}_2\text{O}} \text{R}^1\text{-SR}^2$			
Entry	Zinc reagent 2	Pht-SR 3 ^[a]	Product of type 1 ^[b]
1			

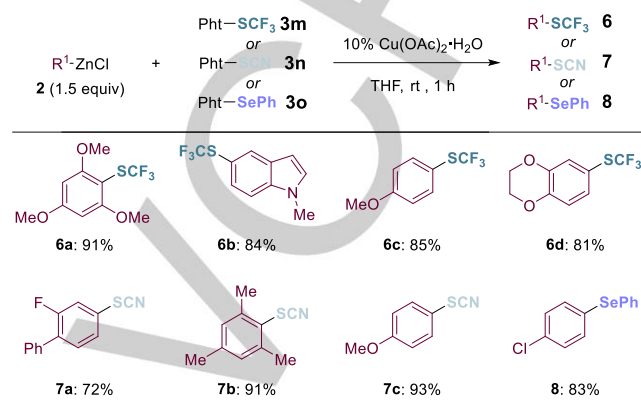
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[a] Pht = phthalimide; [b] Isolated yields of analytically pure products.

Interestingly, this method was further extended to the introduction of a trifluoromethylthio (SCF_3), a thiocyanate (SCN) as well as an phenylselenenyl (SePh) moiety. SCF_3 groups are important bioisosteres to alter the biological activity of drugs, what entails the requirement of elaborated methods for their introduction.^[8] Therefore, commercially available *N*-(trifluoromethylthio)phthalimide **3m** was treated with arylzinc halides of type **2** in the presence of 10% $\text{Cu}(\text{OAc})_2 \cdot \text{H}_2\text{O}$ to give the corresponding

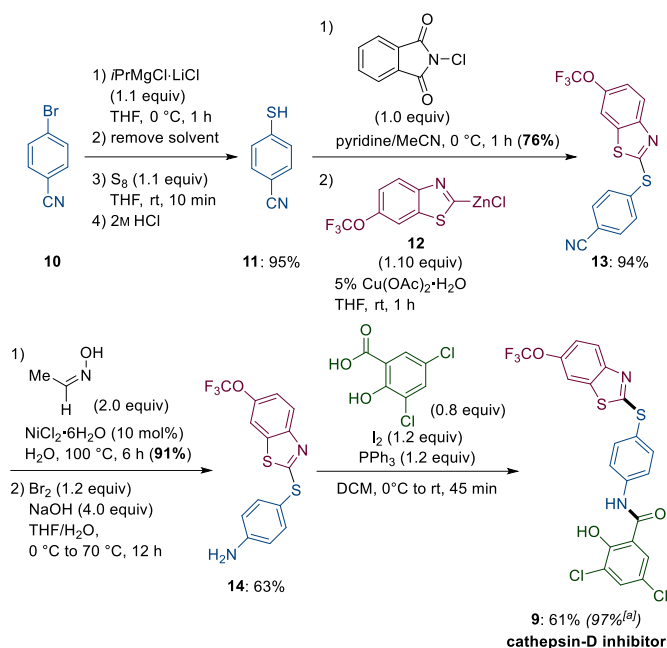
trifluoromethylthiolated products (**6a-d**) in 78-91% yield (Scheme 2). Next, we have extended the scope to thiocyanates. As expected, the reaction of *N*-thiophthalimide **3n**^[7] with arylzinc chlorides of type **2** provided arylthiocyanates **7a-c** in 72-93% yield (Scheme 2). Due to the lower reactivity of these thiophthalimides (**3m-n**), 10% $\text{Cu}(\text{OAc})_2 \cdot \text{H}_2\text{O}$ was necessary.



Scheme 2. Preparation of SCF_3 , SCN and SePh substituted aryl and heteroaryl derivatives via a copper-catalyzed thiolation of arylzinc chlorides.

Additionally, we have extended this method to the preparation of selenium compounds. Thus, selenophthalimide (**3o**) was treated with 4-chlorophenylzinc chloride (**2s**) to give selenoether **8** in 83% yield (Scheme 2). Last, we have applied this methodology to a short synthesis of the bioactive cathepsin-D inhibitor **9**^[9] starting from 4-bromobenzonitrile **10** (Scheme 3). Thus, the conversion of **10** to the corresponding Grignard reagent was achieved with turbo-Grignard ($i\text{PrMgCl} \cdot \text{LiCl}$)^[10], leading after thiolation with elementary sulphur to the thiol **11** in 95% yield.^[11] In this procedure, it was important to remove $i\text{PrBr}$ under *vacuum* after the Br/Mg -exchange step. Using the above described procedure, thiol **11** was converted to the corresponding *N*-thiophthalimide in 76% yield. A subsequent copper-catalyzed thiolation with heteroarylzinc chloride **12**^[7] afforded the desired thioether **13** in 94% yield. Next, the nitrile functionality was hydrolysed using acetaldoxime and $\text{NiCl}_2 \cdot 6\text{H}_2\text{O}$ to provide an intermediate amide in 91% yield.^[12] The corresponding aniline derivative **14** was obtained *via* a Hofmann rearrangement (Br_2 , NaOH) in 63% yield.^[13] Finally, using an iodine/triphenylphosphine-mediated amidation, the desired target **9** was obtained in 61% yield (97% based on recovered starting material).^[14]

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In summary, we have developed a new preparation of thioethers (**1**) from *N*-thiophthalimides **3** via a Cu(OAc)₂·H₂O-catalyzed thiolation of organozinc halides **2**. This method has a broad scope, tolerating various reactive functional groups including aldehydes and nitro groups. An extension to the introduction of SCF₃, SCN and SePh groups was also achieved. The utility of the method was shown in a short synthesis of a bioactive cathepsin-D inhibitor **9**. Further applications are currently underway in our laboratories.

Acknowledgements

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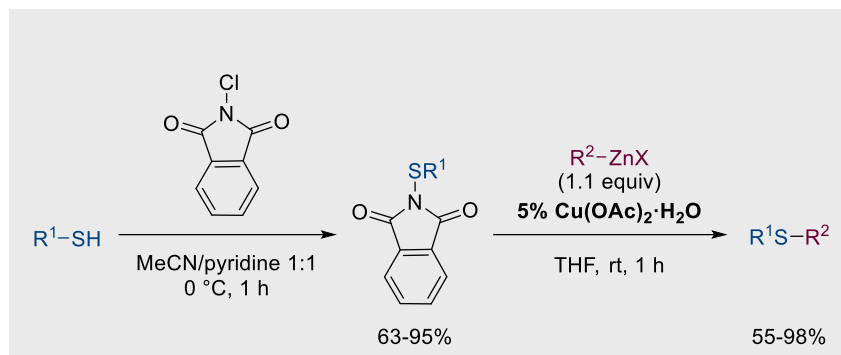
Keywords: electrophilic thiolation • copper-catalysis • organozinc halide • trifluoromethylthiolation • thioether

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